

MTH1001 Mathematical Structures 2025–26, Term 1

Problem Sheet 4

This sheet is for discussion in the week 10 tutorials (Wednesday 26th November). Be ready to discuss (at least) Questions A3, A5(ii), A7(iii), B3 in the tutorial.

Your answers should use mathematical notation correctly and be written in logically correct sentences, including words such as “let”, “if”, “then”. You should introduce any variables that occur in your argument.

SECTION A

These are fairly routine questions. There will be some questions on the January exam of a similar difficulty to these, and some more challenging questions.

A1. The notation $a \mid b$ (“ a divides b ”) for integers a, b is defined to mean that there is some integer m with $b = am$.

In each of the following cases, determine which integers n satisfy the stated condition. Justify your answers.

- | | |
|--------------------|---|
| (i) $n \mid 1$; | (iv) $n \mid 0$; |
| (ii) $1 \mid n$; | (v) $n \mid n$; |
| (iii) $0 \mid n$; | (vi) $a \mid n$ for <i>all</i> integers a . |

A2. **Using the definition of $a \mid b$ (see Question [A1.]),** show that if a, b and c are integers with $a \mid b$ and $b \mid c$ then $a \mid c$.

A3. **Using the definition of $a \mid b$ (see Question [A1.]),** show that if a, b, c and d are integers with $a \mid b$ and $c \mid d$ then $ac \mid bd$.

A4. In each of the following cases, use the Euclidean algorithm to find $g = \gcd(m, n)$ and also find a, b with $am + bn = g$:

- (i) $m = 101, n = 73$, (ii) $m = 801, n = 351$, (iii) $m = 3494, n = 1471$.

Practise on a few examples of your own devising, until you feel totally comfortable with the procedure.

A5. Solve the congruences

- (i) $22x \equiv 29 \pmod{67}$,
(ii) $101x \equiv 172 \pmod{300}$,

(iii) $627x \equiv 37 \pmod{2650}$.

In each case, express your answer in the form $x \equiv a \pmod{n}$ with $0 \leq a < n$.

A6. A *composite* number is an integer $n > 1$ that is not prime. Prove that a composite number n has a prime factor p with $p \leq \sqrt{n}$.

More generally, suppose that n is an integer with k prime factors. Prove that one of these prime factors satisfies $p \leq n^{1/k}$.

A7. By means of unique factorization, or otherwise, prove the following numbers are irrational: (i) $\sqrt{5}$, (ii) $\sqrt{10}$, (iii) $\sqrt[3]{18}$.

SECTION B

These are more challenging questions. There may be questions on the January exam of similar difficulty to these.

- B1. (i) Find the prime factorisation of $10!$, without actually evaluating $10!$.
- (ii) The number $10!$ ends with two zeros (as you can easily check using your calculator). Explain how this fact could have been predicted from your answer to (i).
- (iii) How many zeros are there at the end of the number $17!$?
- (iv) Find the smallest positive integer n for which $n!$ ends in 10 zeros. Justify your answer.

B2. We consider divisibility tests in base-ten notation. Recall that base-ten notation means for instance that

$$7094 = 7 \times 10^3 + 0 \times 10^2 + 9 \times 10 + 4 = 4 \times 1.$$

In general if A is an n -digit number whose digits are $a_{n-1}, a_{n-2}, \dots, a_1, a_0$ from left to right, then

$$A = 10^{n-1}a_{n-1} + 10^{n-2}a_{n-2} + \dots + 10a_1 + a_0 = \sum_{k=0}^{n-1} 10^k a_k.$$

As $10 \equiv 1 \pmod{3}$, then $10^k \equiv 1 \pmod{3}$ and

$$A \equiv \sum_{k=0}^{n-1} a_k = a_{n-1} + a_{n-2} + \dots + a_1 + a_0 \pmod{3}.$$

Therefore A is a multiple of 3 if and only if the sum of its digits $a_{n-1} + \cdots + a_0$ is a multiple of 3.

- (i) Find (and prove) a similar test for divisibility by 9.
- (ii) Prove that A is a multiple of 11 if and only if the *alternating sum* $a_0 - a_1 + a_2 - \cdots + (-1)^{n-1}a_{n-1}$ of its digits is a multiple of 11.

B3. In lectures we proved that the congruence

$$ax \equiv b \pmod{n} \quad (*)$$

can only have a solution if $g \mid b$ where $g = \gcd(a, n)$. Prove that if $g \mid b$ then the integers x solving $(*)$ are exactly the same as the integers solving

$$(a/g)x \equiv (b/g) \pmod{n/g}.$$

Using this, solve the congruence

$$135x \equiv 129 \pmod{222}.$$

B4. Let $m, n \geq 1$ be integers and let $g = \gcd(m, n)$. Define $l = mn/g$. Prove that l is an integer, and that $m \mid l$ and $n \mid l$.

We know there are integers a and b with

$$g = am + bn. \quad (**)$$

Suppose that r is an integer with both $m \mid r$ and $n \mid r$. Using $(**)$ or otherwise, prove that $(mn) \mid (gr)$, and deduce from this that $l \mid r$.

From this, we see that l is a *common multiple* of m and n and that l is a divisor of all common multiples of m and n . Therefore we call l the *least common multiple* of m and n and write $l = \text{lcm}(m, n)$.

Compute $\text{lcm}(98, 175)$.

B5. Let's invent some jargon: a *1-number* is a number ending in 1. More formally a 1-number is $n \in \mathbb{N}$ with $n \equiv 1 \pmod{10}$. The first few 1-numbers are 1, 11, 21, 31, 41, ...

Also define a *1-prime* to be a 1-number n with $n > 1$ and whose only factors which are also 1-numbers are 1 and n . In other words, it is bigger than 1, but not a product of two smaller 1-numbers. So 11, 21, 31, 41, are all 1-primes, but $121 = 11 \times 11$ is a 1-number that's not a 1-prime.

- (i) Find all 1-primes below 500.
- (ii) Prove that every 1-number other than 1 has a 1-prime factor.
- (iii) Prove that every 1-number other than 1 has a factorization into 1-prime factors.
- (iv) (harder) Find a 1-number n with 1-prime factorizations $n = p_1 p_2$ and $n = q_1 q_2$ with $p_1 \neq q_1$ and $p_1 \neq q_2$.